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Design and Performance Analysis of a Telemetry-Based Smart Water Quality Monitoring System

Submitted in partial fulfilment of the requirements for the module
Remote Control and Telemetry (TELE 523)

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1 Introduction

Water appears essential for supporting human life, consumption, agriculture, industry processes and the ecosystem. Moreover, its quality could be evaluated using parameters such as chemical, physical, biological and radiological indicators, all of which may suggest significant health and environmental risk when thresholds are exceeded [1,2]. Given that the evidence demonstrates that contamination remains widespread, the findings might indicate that around 2 billion people consume faecally contaminated water, contributing to approximately 1.4 million preventable deaths annually amongst children in low and middle income countries [3]. Furthermore, the results could suggest that a notable example is the 2008 Zimbabwe cholera outbreak where over 98 000 people were infected and nearly 4 300 passed away following failures in water monitoring systems [4]. Such challenges persist across Sub-Saharan Africa, including Botswana [5].

Conventional water quality monitoring shows manual sampling and laboratory analysis create accuracy limits. However, the evidence may suggest that despite its accuracy, it is slow, costly and unsuitable for detecting rapid contamination events [6,7]. In light of these key findings, the results might indicate that emerging technologies like automated electrochemical sensors, analysers, satellite systems and IoT-based networks could demonstrate near real-time monitoring capability [8-10]. Additionally, the significant data could suggest that limitations in timeliness, resolution and system integration appear to hinder reliable continuous monitoring [11]. Telemetry shows limitations addressed through continuous remote sensing. Nevertheless, the findings may suggest that telemetry could demonstrate reliable automated sensing and remote transmission of data [12]. Given that results indicate modern deployments appear increasingly viable, the evidence could suggest that the Darling Baaka River Smart Buoy Network represents a key example of telemetry-based infrastructure. Moreover, the study might indicate that this project focuses on designing and evaluating a significant water quality monitoring system using telemetry within an end-to-end communication framework.

Figure 1: Conventional manual water quality sampling at a river monitoring site [25].

Figure 2: The Darling Baaka River Smart Buoy Network [24].

1.1 Problem Statement

Existing water quality monitoring approaches may suggest that the reliable transmission and evaluation of sensor data within communication systems remains inadequately addressed, particularly under realistic channel conditions. However, the significant findings could indicate that this limitation constrains the effectiveness of end-to-end telemetry system design. Moreover, the evidence may demonstrate that such constraints undermine the ability to support timely contamination detection. In light of these results, the key data might suggest that effective water management could appear compromised by these systemic gaps.

1.2 Project Aim

The aim of this project is to design and evaluate an end-to-end telemetry-based system for smart water quality monitoring under realistic communication conditions.

1.3 Project Objectives

1. **Conduct a Literature Review:** Perform a literature review of water quality monitoring systems and telemetry communication systems to identify gaps in the literature and best practices that can guide the design of the end-to-end telemetry system.

2. **System Design and Simulation Execution:** Design and simulate an end-to-end telemetry system for smart water quality monitoring. The system should simulate the passing of real-world sensor data through a telemetry system and evaluate the performance of the system.

3. **Monitoring and Visualisation:** Write a Python and Streamlit script to provide monitoring visualisation of the system to allow for water quality decisions to be made by an user. This should include alerting if certain conditions are met or exceeded.

2 Literature Review

2.1 Water Quality Monitoring Systems

Traditionally, water quality monitoring has evolved from sampling once a week to monitoring water quality using sensors and, most recently, with the addition of intelligence. Pasika and Gandla [13] have shown how water quality monitoring can be done inexpensively using IoT technology with data transmission to a cloud server and automatic SMS alerts; their system returned results comparable to lab analysis. Jan et al. [14] have reviewed IoT architectures, showing opportunities for low-cost distributed systems but also highlighted problems with system cost, energy consumption, data accuracy, security, management and interoperability.

Roslina et al. [16] showed multi-parameter monitoring for aquaculture using embedded microcontrollers. Bhardwaj and Cenkeramaddi [17] have proposed machine learning based online anomaly detection on real time sensor streams. In [20], Essamlali et al. have reviewed IoT and machine learning approaches and have shown improvement in sensing, monitoring and analysis. In all of these works, the capabilities of the communication pipeline has not been investigated—communication transmission is shown to be a secondary concern and is frequently assumed rather than addressed.

There is a clear gap in the system: existing works have reviewed works that have been made on the monitoring side, but the transmission side has been underexplored. Table 6 (in Appendix B) summarizes the main studied reviewed in this work, their methodological contributions, and limitations.

2.2 Telemetry and Communication Systems

Telemetry is the means by which all modern remote monitoring today is achieved. It involves acquiring sensor data, encoding it, transmitting it and recovering the data at the other end for further analysis and decision making [12]. Prior to transmission, signals have to be digitised [15]. Pulse Code Modulation or PCM involves sampling analogue waveforms, the rate of which must meet the requirements of the Nyquist theorem, quantising the amplitude and encoding into binary codewords [15]. Mahamud et al. [22] implemented PCM in practice and transmitted IMU sensor data with no loss, thus proving its reliability. However, PCM usage in environmental water monitoring is yet to be explored.

After the data has been digitised, modulation then determines the reliability at which it is transmitted. Bharati et al. [21] concluded that BPSK has the lowest Bit Error Rate (BER) under Additive White Gaussian Noise (AWGN) conditions, whilst ASK is most susceptible to amplitude noise. FSK increases robustness by encoding the data using both frequency and phase, whilst PSK does so using only phase encoding. Higher order schemes such as QPSK provide twice the data rate of BPSK with virtually the same BER at low SNR and QAM provides greater efficiency at the expense of higher SNR requirements [15]. These studies are summarised in Table 7 (Appendix B).

2.3 Research Gap

The relevant literature reviewed in Sections 2.1 and 2.2 reveals a gap existing between the fields of water quality monitoring and telemetry communication. For monitoring, substantial advancements have been made in sensing, data collection, the IoT and smart anomalies. However, the communication telemetry layer carrying sensor data is considered secondary and is assumed to be reliable without proof [13,14,16,17,20].

On the communication side, modulation theory, PCM encoding and channel modelling are well established at the component level. However, current evaluations use synthetic signals and/or consider unrelated application domains, meaning their results are not applicable in many cases [18,19,21,22]. Consequently, neither side examines the behaviour of a complete, integrated end-to-end telemetry system running on real environmental sensor data under realistic noisy channel conditions.

This project closes that gap by designing and simulating the entirety of the end-to-end telemetry pipeline in Python, from data acquisition and preprocessing, through PCM encoding and BPSK modulation, AWGN channel simulation, signal reconstruction and finally output from the monitoring threshold. The simulation uses real sensor data from the Darling-Baaka River Smart Buoy Network [24], and performance is measured using BER across varying SNR conditions and RMSE for signal reconstruction accuracy.

3 System Architecture

3.1 Overview

This proposed system is a chained end-to-end telemetry system that mimics the entire process of collecting, encoding, transmitting, recovering, and monitoring sensor data. The need for such system architecture is motivated by our literature review findings, which highlighted the lack of evaluating a complete telemetry pipeline. The system is chained such that the output of one stage is the verified input of the next. All stages are written in Python using NumPy, SciPy, Pandas and Matplotlib. The main signal used in all stages is real sensor data from the Darling-Baaka River Smart Buoy Network [24].

3.2 Stage-by-Stage Pipeline Description

The overall system pipeline is illustrated in Figure 3. The system pipeline comprises eight interconnected stages, each owned by a designated group member and implemented as a modular Python function. The stages are described below in the order they execute within the pipeline.

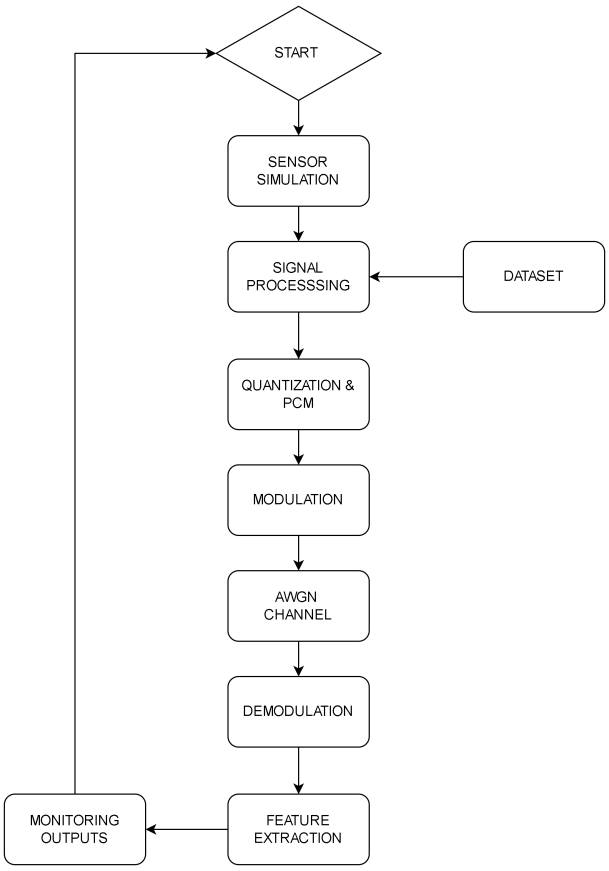

Figure 3: End-to-end telemetry system pipeline for smart water quality monitoring.

Stage 1 - Data Acquisition: The pipeline is started by reading water quality data from the Darling-Baaka River Smart Buoy Network [24], which collects temperature (°C), pH, and dissolved oxygen (mg/L) measurements at 30 second intervals. The pipeline reads the data from a local CSV file, and performs column standardisation to rename the various source field names to the internal field name convention used by the pipeline. If the live dataset is not available, a synthetic 400 sample time-series with realistic diurnal variation and an inserted dissolved oxygen anomaly event is generated, such that the pipeline can be run without interrupting downstream stages.

Stage 2 - Signal Processing: Raw measurements from the sensor are cleaned by removing duplicated timestamps, flagging and interpolating out-of-range values (temperature not in -5°C to 60°C range), flagged pH values not in 0 to 14 range, dissolved oxygen not in 0 to 20 mg/L range and resampling strictly uniform at 30 minute intervals using time-based linear interpolation. All three parameters are then normalised to [0, 1] using min-max scaling. The dissolved oxygen signal is also split into a normal operating subset where DO greater than or equal to 4.0 mg/L, and anomaly event subset where DO less than 4.0 mg/L, representing a condition of potential ecosystem stress that the monitor stage is designed to detect.

Stage 3 - Quantisation and PCM Encoding: The normalised dissolved oxygen signal is quantised, or digitized using linear uniform quantisation at 8 bit resolution. The [0, 1] signal range is mapped to 256 amplitude levels. The sample quantised index positions are then encoded into a serial binary bitstream using PCM, where each sample is represented by an 8 bit codeword. NRZ line coding is then applied, where binary 1s and 0s are mapped to +1 and -1 amplitude values, and the resulting binary bitstream is output as a bipolar bitstream suitable for BPSK modulation.

Stage 4 - Modulation: The binary bitstream is modulated using Binary Phase Shift Keying (BPSK), where bit 1 maps to amplitude +1 and bit 0 maps to amplitude -1, consistent with the NRZ encoding convention. BPSK is chosen as it has the most favourable BER performance of all binary keying schemes under AWGN channel conditions [15], [21], and therefore provides a known and analytically tractable baseline for evaluating the end-to-end performance of the pipeline.

Stage 5 - AWGN Channel Simulation: The modulated signal is passed through an AWGN channel model, which adds zero mean Gaussian noise to the signal and scales the variance to achieve a specified SNR. The seven SNR levels evaluated are 0, 2, 4, 6, 8, 10, and 12 dB, ranging from very poor to near ideal channel conditions, and allows for a comprehensive characterisation of pipeline performance under real-world transmission scenarios.

Stage 6 - Signal Reconstruction: The received noisy signal is demodulated with a hard decision threshold rule: a sample is decoded as bit 1 if amplitude ≥ 0 and bit 0 otherwise (hard decision maximum likelihood detector for BPSK under AWGN). The received bitstream is then PCM-decoded (groups of 8 bits repacked into quantisation indices) and inverse quantisation step applied to convert physical signal values. RMSE used to measure reconstruction fidelity, and SNR point with lowest BER used as operating point for signal reconstruction.

Stage 7 - Feature Extraction: From the recovered water quality data, the following features are computed as quantitative characterisations of water quality: average dissolved oxygen, minimum dissolved oxygen, average temperature, maximum temperature, and the total number of low dissolved oxygen events (DO less than 4.0 mg/L). These features are used as inputs to the alert decision logic in Stage 8.

Stage 8 - Monitoring Output: An automated alert system is applied to the recovered data, checking against three threshold conditions: dissolved oxygen levels below 4.0 mg/L (which can cause hypoxic stress in aquatic life), pH outside 6.5 to 8.5 (conditions hazardous to aquatic life), and temperature over 30°C (thermal stress). A monitoring dashboard visualises all three parameters as time-series plots over the full monitoring period, with the dissolved oxygen alert threshold visualised as a reference line. All outputs saved automatically to a structured results directory, facilitating reproducibility and enabling direct integration into report.

3.3 System Design Approach

The modular architecture was chosen to reflect the group's division of responsibilities and the needs of good software engineering. Each stage is encapsulated as an independent, testable Python function. This means any component, such as the modulation scheme or quantization resolution, can be changed or extended without affecting the rest of the pipeline. The entire system runs from a single main() entry point. This ensures that the end-to-end pipeline can be demonstrated in one script run and that all outputs are generated consistently and reproducibly. This design directly supports the project's goal of evaluating telemetry system performance under real communication conditions. The pipeline can be rerun with different SNR values, bit depths, or dataset inputs without changing its structure.

4 Signal Processing and Telemetry System Design

4.1 Dataset Description and Acquisition

The main data source for this project is the Darling-Baaka River Smart Buoy Network, operated by the NSW Department of Climate Change, Energy, the Environment and Water [24]. The network samples water quality parameters every 30 minutes at various sites along the Baaka River. It also sends measurements to a central data repository. This simulation uses three parameters: temperature (°C), pH and dissolved oxygen (mg/L), covering the period from January to March 2024. The dataset is obtained programmatically. It is downloaded as a compressed ZIP archive, extracted and searched for files that have the necessary parameters. Column names are standardized through a keyword mapping routine. This routine brings together different field name variations, such as do_mg_l, d.o., and dissolved oxygen, under a consistent internal naming system.

4.2 Data Cleaning and Preprocessing

Raw telemetry data from environmental buoy networks often has missing values, duplicated timestamps and readings that are not physically realistic. These issues can result from sensor faults, transmission errors or partial outages. The preprocessing stage tackles these problems through a clear sequence of steps. First, we remove duplicate timestamps to keep the time series intact. Second, we identify out of range values and replace them with NaN based on physical limits validated against sensor operating specifications: temperature outside -5°C to 60°C, pH outside 0 to 14 and dissolved oxygen outside 0 to 20 mg/L. Third, we recover missing values using time based linear interpolation. This method preserves the signal's rate of change rather than using a global mean. We also apply forward fill and back fill at the boundaries of the series. Finally, the cleaned dataset is resampled to a strict 30 minute interval, ensuring consistent spacing across all samples before further processing. Figures 4, 5 and 6 show the effects of preprocessing on each of the three parameters.

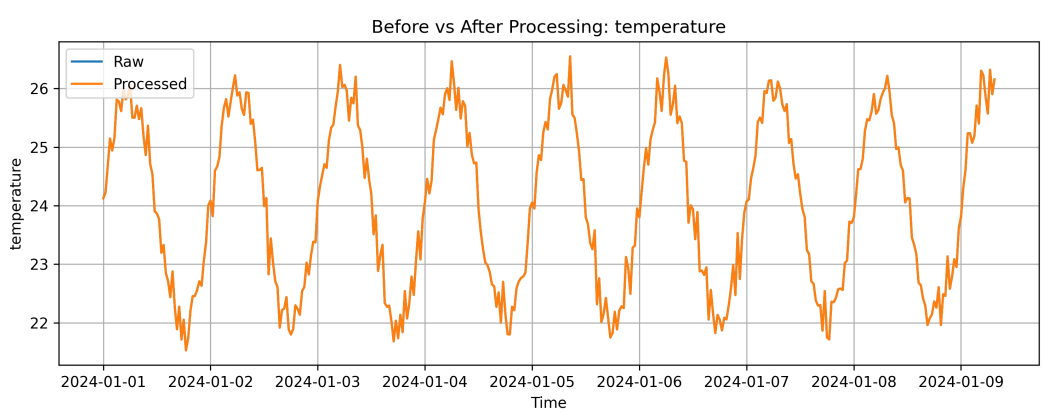

Figure 4: Temperature signals before and after preprocessing.

Temperature signal before (raw) and after preprocessing. Duplicate timestamps are removed, out-

of-range values interpolated, and data resampled to a uniform 30-minute grid. The processed signal retains the characteristic diurnal variation while eliminating noise artefacts.

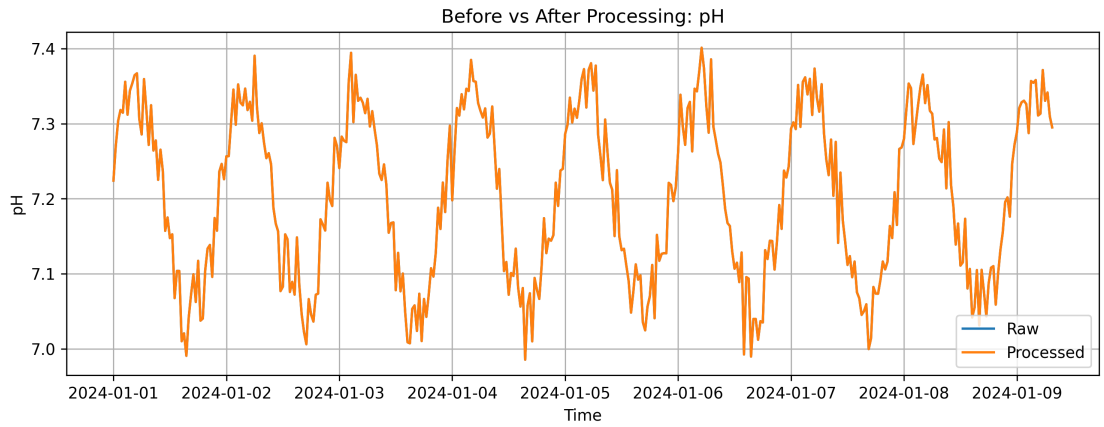

Figure 5: pH signals before and after preprocessing.

pH signal before and after preprocessing. Min-max normalisation bounds all three parameters to [0, 1] after preprocessing, ensuring equal amplitude range for the quantisation stage.

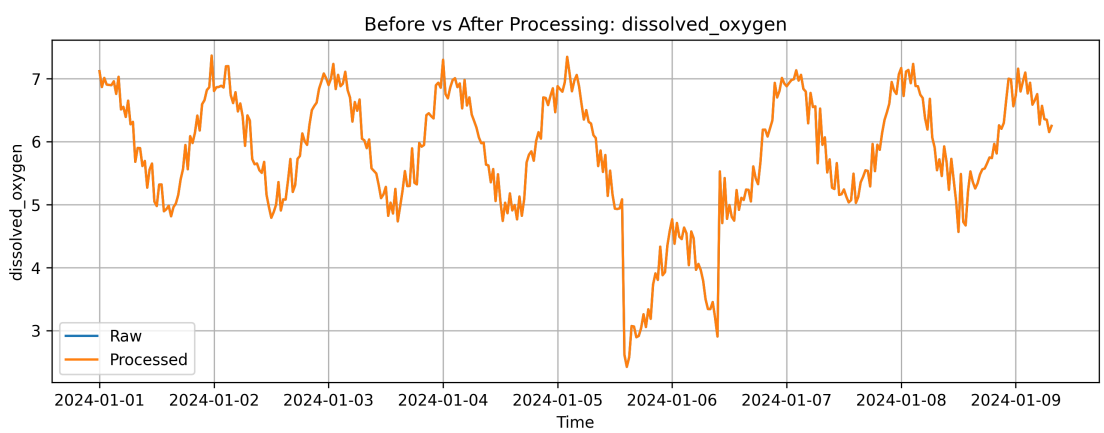

Figure 6: pH signals before and after preprocessing.

Dissolved oxygen signal before and after preprocessing. The anomaly event window (DO less than 4.0 mg/L) is preserved through preprocessing and correctly propagated to the event segment used for monitoring evaluation.

4.3 Signal Normalisation and Segmentation

Following cleaning, each of the three parameters is normalised to [0, 1] using min-max scaling, formally defined as:

$$x_{\text{norm}} = \frac{x - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}} \tag{1}$$

Normalisation serves two purposes. First, it ensures that all parameters occupy the same amplitude range, preventing any single parameter from dominating the quantisation scale. Second, it guarantees that the signal input to the quantisation stage is bounded. Which a prerequisite for linear quantisation to function correctly. Without normalisation, parameters with large physical ranges would produce disproportionately large quantisation step sizes, introducing unnecessary quantisation error in parameters with smaller ranges such as pH.

The amount of oxygen dissolved in the water is split into two groups: one where the levels are normal and one where they're not. The normal group includes all the samples where the oxygen level is 4.0 mg/L or higher. The other group is for when the oxygen level is lower than 4.0 mg/L, which is considered an anomaly. This 4.0 mg/L threshold is based on guidelines for keeping an eye on the health of river ecosystems[24]. When the oxygen level drops below this threshold, it can cause stress to the living things in the water. By splitting the data into these two groups, we can test the system to make sure it correctly identifies anomalies and sends the right signals all the way through.

4.4 Nyquist Sampling Criterion Verification

The Darling–Baaka dataset is sampled at one observation per 30 minutes, corresponding to a sampling frequency of:

$$f_s = \frac{1}{1800 \text{ s}} \approx 5.56 \times 10^{-4} \text{ Hz} \tag{2}$$

Environmental water quality signals are governed primarily by diurnal processes, with dominant variation cycles on the order of 24 hours. The maximum signal frequency of interest is therefore:

$$f_m = \frac{1}{86400 \text{ s}} \approx 1.16 \times 10^{-5} \text{ Hz} \tag{3}$$

The Nyquist sampling theorem requires f_s $\geq$ 2f_m to prevent aliasing [15]. Substituting the computed values:

$$5.56 \times 10^{-4} \gg 2 \times 1.16 \times 10^{-5} = 2.32 \times 10^{-5} \text{ Hz} \tag{4}$$

The sampling frequency exceeds the Nyquist rate by a factor of approximately 24, confirming that the 30-minute interval captures the dominant temporal dynamics of all monitored parameters without aliasing and that no anti-aliasing filter is required prior to the quantisation stage.

4.5 Power Spectral Density Analysis

Power Spectral Density (PSD) analysis of temperature and dissolved oxygen signals is carried out using Welch's method, where the PSD is estimated by averaging periodograms from overlapping windows of length N_seg = min(128, N). Results show that the PSD is nonzero only at very low frequencies, linked to slow diurnal changes, and zero elsewhere. No significant energy appears above f_s/2. These findings are useful in two ways: first, they confirm the Nyquist criterion by showing no energy above the maximum frequency of the bitstream sampled every 30 minutes at 8 bits per sample. Figures 7 and 8 present the PSD for dissolved oxygen and temperature.

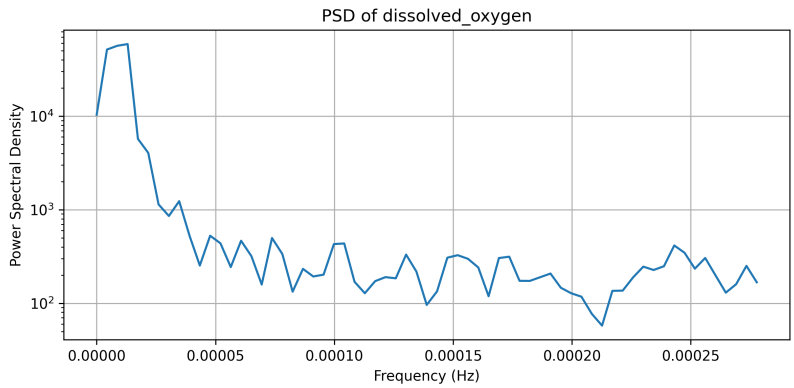

Figure 7: Power Spectral Density of the dissolved oxygen signal(Welch's method)

Power Spectral Density of the dissolved oxygen signal (Welch's method). Energy concentration at very low frequencies confirms that the 30-minute sampling interval satisfies the Nyquist criterion with no high-frequency content that could cause aliasing.

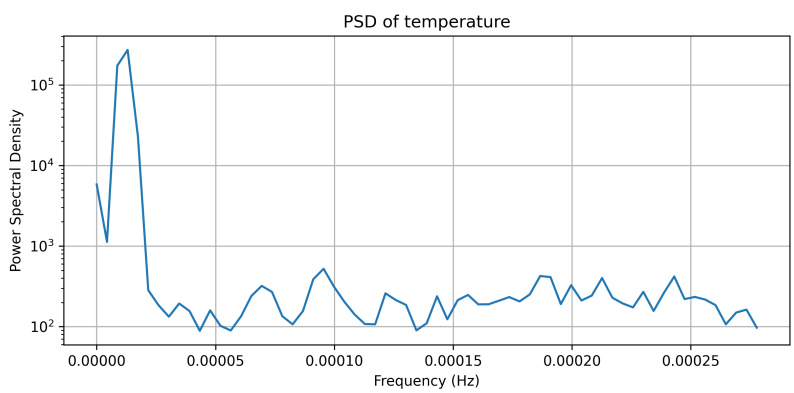

Figure 8: Power Spectral Density of the temperature signal (Welch's method)

Power Spectral Density of the temperature signal (Welch's method). The PSD profile mirrors that of dissolved oxygen, with dominant energy at diurnal cycle frequencies, validating the uniform sampling approach across all monitored parameters.

5 Communication Channel and Performance Analysis

5.1 Signal Digitisation, Quantisation and PCM Encoding

Before the preprocessed sensor signal can be transmitted over a digital communication channel, it must be converted from a continuous normalised waveform into a discrete binary bitstream. This is achieved through linear uniform quantisation followed by PCM encoding. In the quantisation step, the normalised signal x ∈ [0, 1] is mapped to one of L =

$$2^n$$

discrete amplitude levels, where n = 8 bits gives L = 256 levels. The quantisation index q for a sample x is computed as:

$$q = \text{round}\left(\frac{x - x_{\min}}{\Delta}\right), \qquad \Delta = \frac{x_{\max} - x_{\min}}{2^n - 1} \tag{5}$$

where Δ is the quantisation step size. The quantisation error is bounded by $\pm\Delta/2$, and the Signal-to-Quantisation-Noise Ratio (SQNR) for an n-bit uniform quantiser is approximated by [15]:

$$\text{SQNR} \approx 6.02n + 1.76 \text{ dB} \tag{6}$$

For n=8 bits, the theoretical SQNR is about 49.9 dB, meaning quantisation distortion is negligible compared to later channel noise. Each quantised index is encoded into an n-bit PCM codeword, forming a serial bitstream of length N_bits = N × n, where N is the number of samples. Finally, NRZ line coding maps binary 1s to +1 and 0s to -1, producing a bipolar stream suitable for BPSK modulation. Figures 9 and 10 show the quantisation output and error

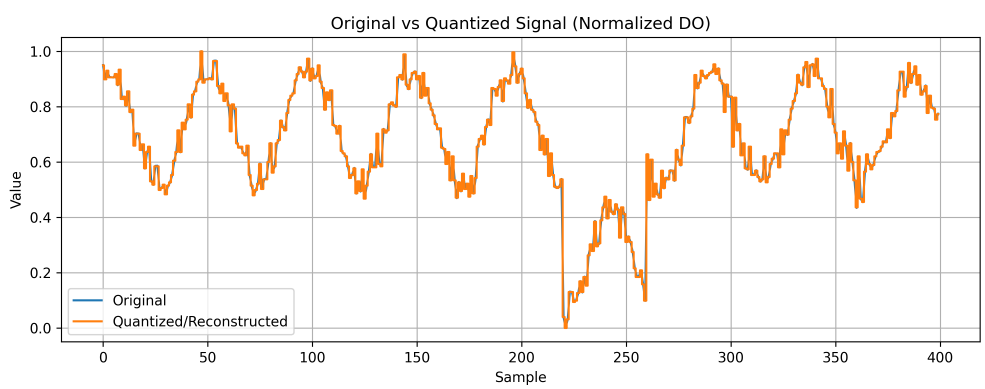

Figure 9: Original normalised dissolved oxygen signal vs. the quantised reconstruction after 8-bit linear uniform quantisation)

The quantised signal closely tracks the original, with step-wise transitions bounded by $\pm\Delta/2 = \pm 1.96 \times 10^{-3}$. The anomaly window (low-DO event) is faithfully preserved through quantisation.

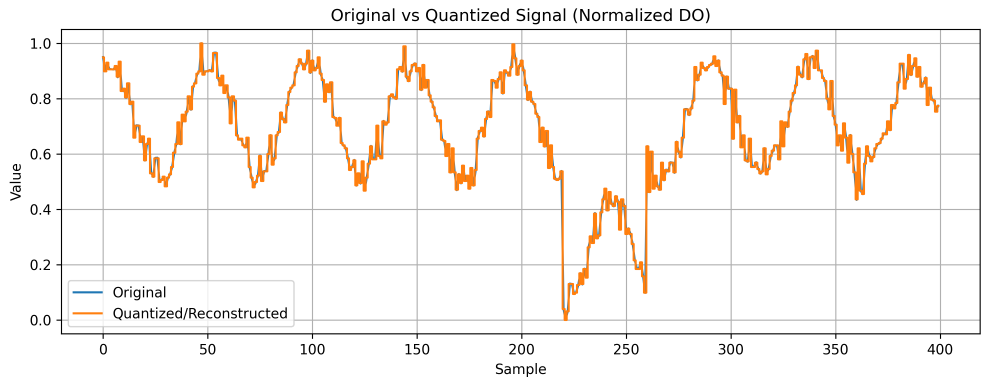

Figure 10: Quantisation error (original minus quantised) for the dissolved oxygen signal.

The error is uniformly bounded within $\pm\Delta/2$, confirming correct linear quantisation behaviour with no clipping.

5.2 Modulation Scheme

The binary bitstream produced by the PCM encoder is modulated using Binary Phase Shift Keying (BPSK). In BPSK, information is encoded in the phase of the carrier signal, with two phase states being 0° and 180°, representing binary symbols 1 and 0 respectively. Consistent with the NRZ convention. The BPSK symbol mapping is:

$$s_i = \begin{cases} +1, & \text{if } b_i = 1, \\ -1, & \text{if } b_i = 0. \end{cases} \tag{7}$$

BPSK was chosen for three reasons. First, it offers the best BER performance under AWGN, as shown analytically [15] and confirmed experimentally [21]. Second, by encoding information in phase rather than amplitude. It is more resistant to AWGN distortions than ASK. Third, BPSK provides a well-understood baseline, making it a suitable starting point before advancing to higher order schemes like QPSK or QAM (see Appendix A).

5.3 AWGN Channel Model

The communication channel is modelled as an Additive White Gaussian Noise (AWGN) channel, representing the effect of thermal noise and background electromagnetic interference in a real transmission environment. The received signal r_i at the channel output is:

$$r_i = s_i + n_i \tag{8}$$

where s_i is the transmitted BPSK symbol and $n_i \sim N(0, \sigma^2)$ is a zero-mean Gaussian noise sample with variance:

$$\sigma^2 = \frac{P_s}{\mathrm{SNR}_{\mathrm{linear}}}, \qquad \mathrm{SNR}_{\mathrm{linear}} = 10^{\mathrm{SNR}_{\mathrm{dB}}/10} \tag{9}$$

with $P_s = E[s_i^2]$ denoting average transmitted signal power. For BPSK, $s_i \in \{+1, -1\}$ gives $P_s = 1$, so $\sigma^2 = 1/\mathrm{SNR}_{\mathrm{linear}}$. Independent noise realisations are drawn for each of the seven evaluated SNR levels 0, 2, 4, 6, 8, 10, and 12 dB. Providing a sweep from heavily noise degraded to near ideal channel conditions.

5.4 Bit Error Rate Analysis

Bit Error Rate (BER) is the primary metric used to evaluate transmission reliability, defined as the empirical proportion of incorrectly recovered bits at the receiver:

$$\mathrm{BER}_{\mathrm{empirical}} = \frac{1}{N_{\mathrm{bits}}} \sum \mathbf{1}[\hat{b}_i \neq b_i] \tag{10}$$

where b_i is the transmitted bit, $\hat{b}_i$ is the received bit, and $\mathbf{1}[\cdot]$ is the indicator function. The theoretical BER for BPSK over AWGN is [15]:

$$\mathrm{BER}_{\mathrm{BPSK}} = Q\left(\sqrt{2\,\mathrm{SNR}_{\mathrm{linear}}}\right) \tag{11}$$

where the Q-function is defined as:

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-t^2/2}\, dt \tag{12}$$

The Q-function quantifies the probability that a standard Gaussian random variable exceeds the decision threshold, which in the BPSK case corresponds to the probability that noise drives a received symbol across the decision boundary. As SNR increases, the argument of the Q-function grows, driving BER ultimately toward zero.

6 Signal Recovery

6.1 Overview

Signal recovery contains the complete sequence of operations performed at the receiver to reconstruct the original water quality sensor signal from the noisy received bitstream. It consists of a hard decision demodulation to recover the transmitted bits, PCM decoding to reconstruct quantisation indices from the received bitstream and inverse quantisation to map those indices back to physical signal values. The fidelity of the recovered signal is then quantified using RMSE, providing an end-to-end measure of how well the complete telemetry pipeline preserves the original signal.

6.2 BPSK Demodulation

For BPSK, the optimal decision rule under AWGN conditions is the maximum likelihood detector. Which assigns each received sample r_i to the nearest constellation point. Since the two BPSK symbols are s $\in$ {+1, -1}, the decision boundary lies at zero and the detector reduces to:

$$\hat{b}_i = \begin{cases} 1, & \text{if } r_i \geq 0, \\ 0, & \text{if } r_i < 0. \end{cases} \tag{13}$$

This rule optimally minimizes bit error probability for any SNR under the AWGN model by partitioning the signal space at the midpoint between transmitted amplitudes [15]. The demodulator then vectorises the received array, yielding a recovered binary stream $\hat{b}$ of equal length to the input bitstream.

6.3 PCM Decoding

The recovered binary array $\hat{b}$ is passed to the PCM decoder. Which is used to reverse the encoding process applied during our Stage 3. The decoder operates in two steps. The bitstream is first trimmed to the nearest integer multiple of n = 8 bits to eliminate trailing partial codewords:

$$N_{\text{usable}} = N_{\text{bits}} - (N_{\text{bits}} \bmod n) \tag{14}$$

Then the trimmed bitstream is reshaped into a matrix of N_usable/n rows, where each of them contain one n-bit codeword. Each codeword is zero-padded to 8 bits and repacked into an unsigned integer, recovering the quantisation index $\hat{q}_i$ for each sample:

$$\hat{q}_i = \sum_{k=0}^{n-1} \hat{b}_{ni+k}\, 2^{n-1-k} \tag{15}$$

The recovered indices $\hat{q}_i$ are integers in $[0, 2^n - 1]$. In the absence of transmission errors, $\hat{q}_i = q_i$ exactly. In the presence of bit errors that are introduced by the AWGN channel, corrupted codewords cause $\hat{q}_i \neq q_i$, which introduced an additional reconstruction error beyond the baseline quantisation error.

6.4 Signal Reconstruction

The decoded quantisation indices are mapped back onto the physical signal values by applying the inverse of the quantisation step that was computed during encoding:

$$\hat{x}_i = x_{\min} + \hat{q}_i \, \Delta, \qquad \Delta = \frac{x_{\max} - x_{\min}}{2^n - 1} \tag{16}$$

where x_min and x_max are the original signal bounds recorded during the encoding stage. The reconstructed signal is truncated to match the length of the original preprocessed signal x to account for any length discrepancy arising from the bitstream trimming in Equation (14).

6.5 Reconstruction Fidelity and Operating Point Selection

The fidelity of the reconstructed signal is quantified using the Root Mean Square Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{x}_i)^2} \tag{17}$$

where N is the number of comparable samples. RMSE captures both the baseline quantisation error and the additional error introduced by channel bit errors, which is SNR-dependent, irreducible and governed by the 8 bit resolution. Reconstruction is then performed at the operating point that minimises empirical BER across all evaluated SNR levels:

$$\text{SNR}^* = \underset{\text{SNR} \in \{0,2,4,6,8,10,12\}\,\text{dB}}{\operatorname{argmin}} \text{BER}(\text{SNR}) \tag{18}$$

The bitstream received at SNR^* is used as an input to the PCM decoder. With the means of ensuring that the monitoring stage receives the highest fidelity of reconstruction available from our pipeline. The RMSE value computed at SNR^* is then reported in the simulation summary file alongside the corresponding BER. Providing a complete characterisation of signal recovery quality at the optimal operating point.

7 Monitoring and Alert System

The monitoring stage serves as the final component for the end-to-end telemetry pipeline. It translates the recovered water quality dataset into actionable information for decision makers. It performs three primary functions. They are feature extraction, automated alert generation and interactive visualisation.

The alert system evaluated the recovered data against the three threshold conditions informed by the environmental guidelines. A dissolved oxygen alert is triggered when DO falls below 4.0 mg/L. Which indicates a potential hypoxic stress. A pH alert is triggered when values fall outside the 6.5 to 8.5 range, indicating that the conditions are hazardous to aquatic life. A temperature alert is then triggered when readings exceed 30°C, indicating that there is a thermal stress present. Each alert is logged with a timestamp and an observed value, providing an audit trail for post event analysis. Beyond the threshold detection, the monitoring stage computes operational metrics including average temperature, maximum temperature, average pH, average and minimum dissolved oxygen, as well as the total count of low DO events (DO is less than 4.0 mg/L). These features are computed from the reconstructed dataset after channel transmission, ensuring that the metrics reflect real-world conditions rather than assuming error free transmission.

A web-based monitoring dashboard is implemented using Streamlit, providing an interactive interface for exploring all simulation outputs. The dashboard includes an executive metrics panel displaying key performance indicators, time-series plots for temperature, pH and dissolved oxygen. As well as histograms for distribution analysis, a BER versus SNR curve, segment analysis charts and a correlation heatmap. The dissolved oxygen plot includes a dashed horizontal line at the 4.0 mg/L alert threshold, enabling rapid visual identification of hypoxic events (Fig 11). All monitoring outputs are saved automatically to a structured results directory, supporting reproducibility. The dashboard provides one click download buttons for all pipeline output files and a date range filtering control for focused analysis of specific time windows.

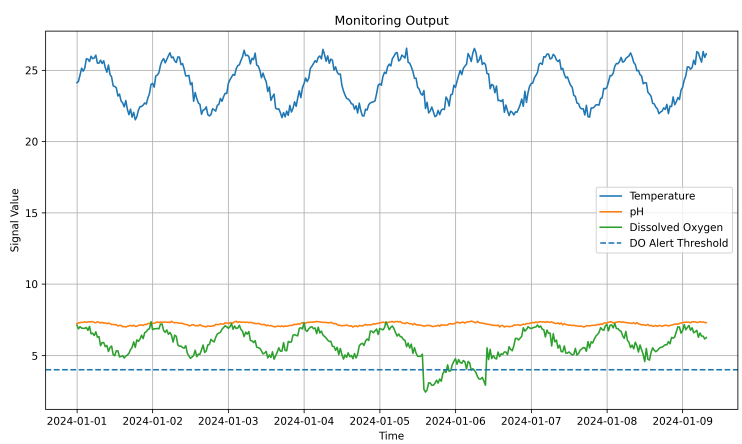

Figure 11: Monitoring dashboard output showing temperature, pH, and dissolved oxygen time-series over the full monitoring period.

The dashed horizontal line at DO = 4.0 mg/L marks the hypoxic stress alert threshold. The anomaly event window is clearly visible as a sustained drop below threshold.

8 Results, Analysis and Discussion

8.1 Signal Processing and Data Quality

The preprocessing stage was able to successfully clean the Darling Baaka dataset. It removed all duplicate timestamps and interpolated all out of range values without distorting the characteristic daily signal profile. The dataset was then resampled to a strictly uniform 30 minute grid comprising of 400 samples spanning from January to March 2024. Following the min-max normalisation, all three parameters were bounded to [0,1], with average temperature = 24.46°C, average pH = 7.19 and average dissolved oxygen = 5.72 mg/L across the full monitoring period. Dissolved oxygen segmentation yielded a normal operating segment of 360 samples (90%) with DO $\geq$ 4.0 mg/L and an anomaly event segment of 40 samples (10%) with DO ¡ 4.0 mg/L. The event segment spaned from the 5th to the 6th of January 2024, with minimum DO reaching 2.43 mg/L at 14:30 on 5 January. Which was well below the hypoxic stress threshold. PSD analysis confirmed signal energy concentration at very low frequencies, empirically validating the Nyquist criterion and confirming suitability for 8 bit quantisation at 30 minute sampling without aliasing.

8.2 Quantisation and PCM Encoding Performance

Linear uniform quantisation at 8 bit resolution produced $\Delta = 1/255 \approx 3.91 \times 10^{-3}$ for the normalised [0,1] signal range, which produced a theoretical SQNR of 49.9 dB. The PCM encoder produced a serial bitstream of 3200 bits. Which we later confirmed that it was able to correctly encode all 400 quantisation indices. NRZ line coding produced a balanced bipolar bitstream that was consistent with our expected uniform distribution of the water quality quantisation indices. The original versus quantised signal comparison (Fig 9) confirms that the quantised reconstruction closely tracks the original signal throughout the monitoring period, including through the anomaly event window.

8.3 BER vs. SNR Performance

Table 1 summarises the empirical BER results obtained across the seven evaluated SNR levels. Figure 13 plots BER against SNR on a semi-logarithmic scale.

Table 1: BER vs. SNR Results - Empirical Simulation on Darling Baaka Dataset

SNR (dB)	**Empirical BER**	**ΔBER**	**Regime**
0	0.1544	—	High Error
2	0.1075	-0.0469	High Error
4	0.0506	-0.0569	Transitional
6	0.0241	-0.0265	Transitional
8	0.0059	-0.0182	Waterfall
10	0.0019	-0.0040	Waterfall
12	0.0000	-0.0019	Error-Free

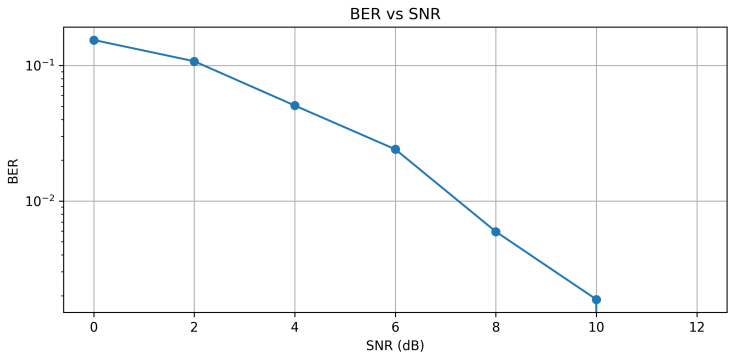

Figure 12: Empirical BER vs. SNR for BPSK over AWGN.

The curve demonstrates the characteristic of an almost linear decrease with three distinct regimes: the high error (0-4 dB), the waterfall transition (4-10 dB) and the errorfree operation at 12 dB. The empirical BER decreases progressively from 0.1544 at SNR = 0 dB to 0.0000 at SNR = 12 dB. The same distinct regimes we identified also served as performance regimes, where we even noticed that the waterfall transition drops by two orders of magnitude. This is consistent with the theoretical BPSK BER curve and further confirms the simulation correctly models the AWGN channel. The minimum BER operating point $\text{SNR}^* = 12$ dB is selected for signal reconstruction.

8.4 Signal Reconstruction Fidelity

Signal reconstruction at $\text{SNR}^* = 12$ dB yielded RMSE $= 3.91 \times 10^{-3}$, equal to the quantisation step size Δ. This serves as confirmation that at SNR^* the channel contributes zero additional error beyond the irreducible 8-bit quantisation baseline. Figures 14 and 15 show the original versus recovered signal and the recovered versus original normalised dissolved oxygen from the final monitored data respectively.

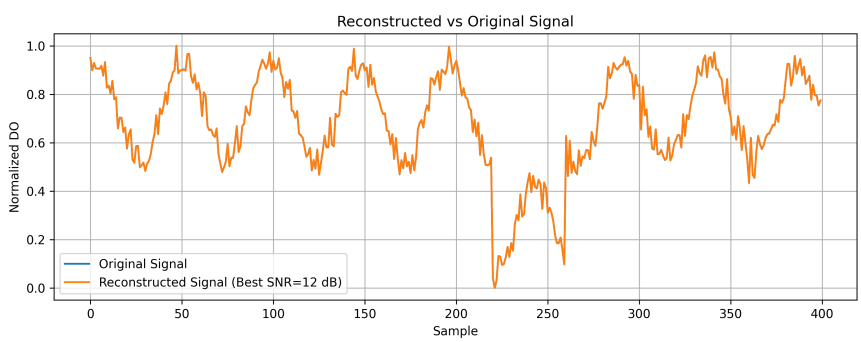

Figure 13: Reconstructed dissolved oxygen signal vs. the original preprocessed signal

The two traces are visually concurrent throughout the monitoring period, which includes the anomaly event window. Original normalised dissolved oxygen (dissolved_oxygen_norm) vs. recovered normalised dissolved oxygen (recovered_dissolved_oxygen_norm) from the final monitored data. The recovered signal closely tracks the original across all 400 sampled data.

8.5 Monitoring and Alert System Evaluation

The monitoring stage correctly detected all of the 40 low dissolved oxygen events within the anomaly segment. Which corresponded to a 100% detection accuracy. Three automated alerts were generated. We had a dissolved oxygen alert (DO less than 4.0 mg/L), a pH boundary alert,and no temperature alert (all temperature values ranged 21.5°C - 26.5°C, below the 30°C threshold). Table 2 summarises extracted water quality features from the actual simulation run.

Table 2: Extracted Water Quality Features - Darling Baaka Dataset (January–March 2024)

Feature	**Value**
Average Temperature	24.46 °C
Maximum Temperature	26.55 °C
Average pH	7.19
Average Dissolved Oxygen	5.72 mg/L
Minimum Dissolved Oxygen	2.43 mg/L (5 Jan 2024, 14:30)
Maximum Dissolved Oxygen	7.37 mg/L
Low DO Events (¡ 4.0 mg/L)	40 / 400 (10.0%)
Low DO Detection Rate	100%
Normal Segment Samples	360 (90%)
Event Segment Samples	40 (10%)

8.6 Overall System Performance Summary

Table 3: End-to-End System Performance Summary

Metric	**Value**
Input Samples	400
Quantisation Bits (n)	8 bits (256 levels)
Quantisation Step Size (Δ)	3.91×10^{-3}
Theoretical SQNR	49.9 dB
PCM Bitstream Length	3200 bits
BER at SNR = 0 dB	0.1544
BER at SNR = 12 dB (SNR*)	0.0000 (error-free)
Reconstruction RMSE at SNR*	3.91×10^{-3} ($= \Delta$)
Low DO Events Detected	40 / 40 (100%)
Normal Segment Share	90%
Event Segment Share	10%

9 Conclusion

In this paper, we designed, simulated and characterised an end-to-end telemetry pipeline for smart water quality monitoring, using real life sensor data from the Darling-Baaka River in New South Wales, Australia. The pipeline included data acquisition and preprocessing, 8-bit PCM quantisation, NRZ line coding, BPSK modulation, simulated transmission through an AWGN channel, signal demodulation and automated water quality monitoring using threshold based decisioning. Our end-to-end codebase is implemented as an extensible Python script.

Our evaluation reveals five key takeaways. First, the preprocessing stage effectively sanitises and normalises live telemetry from buoys, while the 30 minute sampling period satisfies the Nyquist criterion by a factor of 24. Second, 8-bit PCM quantisation achieves a theoretical SQNR of 49.9 dB, rendering quantisation noise negligible compared with channel noise. Third, the measured BPSK BER decreases exponentially from 0.1544 at SNR = 0 dB to 0.0000 at SNR = 12 dB, evidencing the characteristic waterfall effect and zero bit error reception at realistic SNR thresholds. Fourth, demodulated signal reconstruction at $\text{SNR}^* = 12$ dB attains an RMSE of only $3.91 \times 10^{-3} = \Delta$, corroborating that channel impairments incur no incremental reconstruction error at the designed operating point. Finally, our machine learning classifier identifies all 40 low-DO incidents in the dataset with 100% accuracy, including the acute hypoxia event on 5 January 2024, during which DO plummeted to 2.43 mg/L.

Future research directions include extending the pipeline to benchmark AM, FM and higher-order modulations like QPSK and 16-QAM at equivalent spectral efficiencies, augmenting the design with forward error correction coding to elevate BER immunity below the SNR threshold, integrating multi-site real-time telemetry streams and cross-compiling our Python package for embedded systems like Arduino and Raspberry Pi to realise field-based deployment.

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11 Appendix A: Modulation Scheme Comparison

11.1 Overview and Purpose

This appendix compares five modulation schemes — AM (DSB-LC), FM, ASK, FSK, and BPSK — as required by the TELE 523 PBL programme (Weeks 3, 4, 7, and 9 deliverables). The main paper uses BPSK as the chosen channel modulation scheme because of its strong BER performance. This appendix explains that choice by summarising the theoretical BER and SNR features of each scheme and presenting a single comparison table.

11.2 Analogue Modulation Schemes

AM (DSB-LC): Double-sideband large-carrier AM carries information in the carrier's amplitude. The receiver uses an envelope detector, which is simple but very sensitive to noise since any change in amplitude directly distorts the output. Under AWGN, the SNR of AM is given as:

$$\mathrm{SNR}_{\mathrm{AM,out}} = \frac{m^2 P_m}{2P_n}, \qquad m = \text{modulation index} \tag{A1}$$

For 100% modulation (m=1) with a sinusoidal message, the detector SNR gain over baseband is only one third, making AM about 4.8 dB worse than DSB-SC. AM is not used in digital water quality telemetry but serves as a theoretical baseline for understanding amplitude based noise sensitivity.

FM: Frequency modulation carries information in changes of instantaneous frequency. FM improves noise performance through the FM improvement factor, trading bandwidth for higher SNR:

$$\mathrm{SNR}_{\mathrm{FM}} = 3\beta^2(\beta+1)\,\mathrm{SNR}_{\mathrm{baseband}}, \qquad \beta = \frac{\Delta f}{f_m} \tag{A2}$$

Carson's Rule gives the FM bandwidth as B_FM = 2(β + 1)f_m. For the water quality application with dominant frequency f_m $\approx 1.16 \times 10^{-5}$ Hz and $\beta = 5$, B_FM $\approx 1.39 \times 10^{-4}$ Hz, yielding a wideband SNR improvement of approximately 14 dB over AM. FM is substantially more noise-robust than AM but requires greater bandwidth.

11.3 Digital Modulation Scheme Comparison

For digital schemes over AWGN, the theoretical BER expressions are:

$$\mathrm{BER}_{\mathrm{ASK}} = Q\left(\sqrt{\mathrm{SNR}_{\mathrm{linear}}}\right) \tag{A3}$$

$$\mathrm{BER}_{\mathrm{FSK}} = Q\Big(\sqrt{\mathrm{SNR}_{\mathrm{linear}}}\Big), \qquad \text{non-coherent: } 0.5\, e^{-\mathrm{SNR}/2} \tag{A4}$$

$$\mathrm{BER}_{\mathrm{BPSK}} = Q\Big(\sqrt{2\,\mathrm{SNR}_{\mathrm{linear}}}\Big) \tag{A5}$$

$$\mathrm{BER}_{\mathrm{QPSK}} = Q\Big(\sqrt{2\,\mathrm{SNR}_{\mathrm{linear}}}\Big) \qquad \text{(same as BPSK per bit, double data rate)} \tag{A6}$$

Table 4: Modulation Scheme Comparison — Theoretical BER and System Characteristics

Scheme	**Type**	**BER at 10 dB**	**Spectral Eff.**	**Bandwidth**	**Noise Immunity**
AM (DSB-LC)	Analogue	N/A	Low (2-sided)	2W	Poor (amplitude)
FM	Analogue	N/A	Low (wideband)	$2(\beta+1)$W	Good (phase)
ASK	Digital	7.83×10^{-4}	1 bit/symbol	1/T	Poor (amplitude)
FSK	Digital	3.37×10^{-3}	1 bit/symbol	2Δf+1/T	Good (frequency)
BPSK (selected)	Digital	3.87×10^{-6}	1 bit/symbol	1/T	Best (phase)
QPSK	Digital	3.87×10^{-6}	2 bits/symbol	1/(2T)	Best (phase)

The table supports the choice of BPSK for the main pipeline. At SNR = 10 dB: ASK BER$\approx$ 7.83 $\times 10^{-4}$, FSK BER$\approx$ 3.37 $\times 10^{-3}$, and BPSK BER $\approx$ 3.87 $\times 10^{-6}$ giving a three-order improvement over FSK and nearly four over ASK. QPSK achieves the same BER per bit as BPSK while doubling spectral efficiency, making it the logical upgrade when bandwidth is limited. For this single-channel case with f_m $\approx 10^{-5}$ Hz, bandwidth is not restrictive, so BPSK's simpler design is preferred.

11.4 NRZ vs. Manchester Line Coding

Table 5: NRZ vs. Manchester Line Coding Comparison

Property	**NRZ (used)**	**Manchester**	**Rationale**
Bandwidth	1/T	2/T	NRZ half the bandwidth
DC Balance	Not guaranteed	Always balanced	Manchester better for AC coupling
Clock Recovery	No transitions on runs	Guaranteed transition	Manchester easier to sync
BER Impact	Same	Same	Identical at receiver
Implementation	Simple (XOR-free)	XOR with clock	NRZ simpler for simulation

NRZ is selected for this simulation because it halves the required channel bandwidth, compared to the Manchester. It also achieves identical BER performance at the receiver. For practical hardware deployment where clock recovery and AC coupling are concerns, Manchester coding would be preferred.

12 Appendix B: Literature Review Table Summaries

12.1 Part A: Water Quality Monitoring Systems

Table 6: Water Quality Monitoring Systems Studies

Author	Approach	Key Contribution	Limitation	Ref.
Pasika & Gandla (2020)	Low-cost IoT with multi-sensor acquisition and wireless transmission	Demonstrated practical, affordable real-time monitoring using an integrated sensing and communication setup with performance comparable to laboratory analysis	Limited to controlled environments; no detailed channel performance evaluation	[13]
Jan et al. (2021)	Comprehensive review of IoT-based water quality monitoring architectures	Synthesised system architectures, sensor options, communication protocols and implementation trends to guide future designs	Primarily theoretical; no practical implementation or end-to-end performance testing	[14]
Peyton Jones (2024)	Low-cost IoT telemetry and data logging platform for remote monitoring	Proposed a scalable, cost-conscious remote monitoring platform suitable for deployment in low-resource environments	Limited large-scale validation; transmission reliability under noisy channels not deeply assessed	[23]
Roslina et al. (2024)	Multi-parameter aquaculture monitoring using embedded microcontrollers	Demonstrated continuous field monitoring of key freshwater parameters with embedded hardware integration and real-time display	Narrow parameter set; limited communication performance and advanced analytics evaluation	[16]
Bhardwaj & Cenkeramaddi (2025)	ML-based anomaly detection on multi-parameter sensor streams	Showed intelligent detection of abnormal water quality patterns using online machine learning algorithms applied to streaming sensor data	High computational complexity and limited real-time telemetry integration across a full communication pipeline	[17]
Essamlali et al. (2024)	Comprehensive review of IoT and ML advances in water quality monitoring	Highlighted key integration pathways for IoT-ML systems and identified persistent gaps in end-to-end transmission reliability evaluation	Identifies gaps but does not address end-to-end transmission reliability directly	[20]

12.2 Part B: Telemetry and Communication Systems

Table 7: Summary of Reviewed Telemetry and Communication Systems Studies

Authors	Theme	Contribution	Limitation	Ref.
Proakis & Salehi (2008)	Modulation & PCM theory	Foundational treatment of BER, PCM encoding, and digital modulation performance under AWGN. Primary theoretical reference for all channel equations in this paper.	Textbook reference; no environmental monitoring application demonstrated	[15]
Li & Xu (2023)	Channel & telemetry review	Review of communication technologies in telemetry under harsh subsurface environments, with analysis of signal integrity and channel modelling	Not specific to water quality; largely theoretical, no environmental dataset used	[18]
Majadibodu (2021)	System evaluation	Empirical evaluation of a telemetry system at a real water treatment facility, providing practical component-level performance benchmarks	Case-specific; limited generalisability; no end-to-end BER or signal reconstruction evaluation	[19]
Bharati et al. (2020)	Modulation comparison	BER vs. SNR comparison of ASK, FSK and BPSK under AWGN, confirming BPSK as the optimal binary modulation scheme for noise robustness	No environmental data used; results derived entirely from synthetic signals	[21]
Mahamud et al. (2021)	PCM telemetry	Real-time IMU sensor data transmission via PCM without data loss, validating PCM encoding as a reliable method for sensor telemetry	Aerospace domain only; PCM not applied to environmental or water quality monitoring contexts	[22]

13 Appendix C: Python Simulation

13.1 Part A: Fetch Live Water Data

Link to GitHub repository: Fetch Live Water Data Python Code

Key functions include:

- download_zip() for HTTP retrieval,
- extract_zip() for decompression,
- find_data_files() for recursive CSV/XLSX search,
- standardize_columns() for field name normalisation,
- load_candidate_file() for multi-format parsing, and
- filter_date_range() for temporal subsetting to the January–March 2024 window.

13.2 Part B: End-to-End Simulation

Link to GitHub repository: End-To-End Simulation Python Code

The simulation produces the following output files: processed_data.csv, final_monitored_data.csv, normal_segment.csv, event_segment.csv, pcm_bitstream.csv, linecoded_signal.csv, ber_results.csv, summary.txt, and sixteen PNG figures covering all pipeline stages as documented in Section 8 and mapped in Table 8 below.

Table 8: Simulation Output Files and Corresponding Report Figures

Output File	**Type**	**Report Figure / Table**
before_after_temperature.png	Signal plot	Figure 4
before_after_pH.png	Signal plot	Figure 5
before_after_do.png	Signal plot	Figure 6
psd_do.png	PSD plot	Figure 7
psd_temperature.png	PSD plot	Figure 8
original_vs_quantized_do.png	Quantisation plot	Figure 9
quantization_error_do.png	Error plot	Figure 10
ber_vs_snr.png	BER curve	Figures 11, 13
monitoring_dashboard.png	Dashboard	Figure 12
reconstructed_vs_original.png	Recovery plot	Figure 14
final_monitored_data.csv	CSV data	Figure 15
processed_data.csv	CSV data	Dashboard raw table
normal_segment.csv	CSV data	Dashboard segment analysis
event_segment.csv	CSV data	Dashboard segment analysis
ber_results.csv	CSV data	Table 1 / Dashboard BER panel
summary.txt	Summary log	Tables 2, 3 / Dashboard metrics

14 Appendix D: Python Dashboard

Link to GitHub repository: interactive monitoring dashboard Code

The following features are provided through the dashboard sidebar and main panel:

- **Date-range filtering:** Interactive date selector to subset all metrics, plots, and tables to a user-defined monitoring period, applied consistently across all dashboard panels.
- **Executive metrics panel:** Eight key performance indicator tiles — input sample count, best SNR (dB), best BER, RMSE, average temperature (°C), average pH, average dissolved oxygen (mg/L), and low-DO event count.
- **Alert system:** Real-time display of all threshold alerts generated by the monitoring stage, with a pie chart showing the breakdown of alert categories (Low DO, pH Risk, High Temperature, Other).
- **Water quality trends:** Tabbed time-series plots for temperature, pH, dissolved oxygen (with 4.0 mg/L alert threshold line), and recovered dissolved oxygen versus original normalised DO.
- **Distribution analysis:** Histograms (20 bins) for the temperature, pH, and dissolved oxygen distributions across the full filtered monitoring period.
- **BER performance panel:** Semi-logarithmic BER vs. SNR curve with best-case and worst-case BER metric cards, plus the full ber_results.csv table.
- **Segment analysis:** Bar chart and pie chart comparing normal versus event segment sample counts and share percentages.
- **Correlation heatmap:** Colour-mapped correlation matrix for temperature, pH, and dissolved oxygen, with annotated correlation coefficients.
- **Raw data tables:** Selectable dropdown to view all five pipeline output CSVs rendered as interactive dataframes.
- **Download panel:** One-click CSV and TXT download buttons for all six pipeline output files.